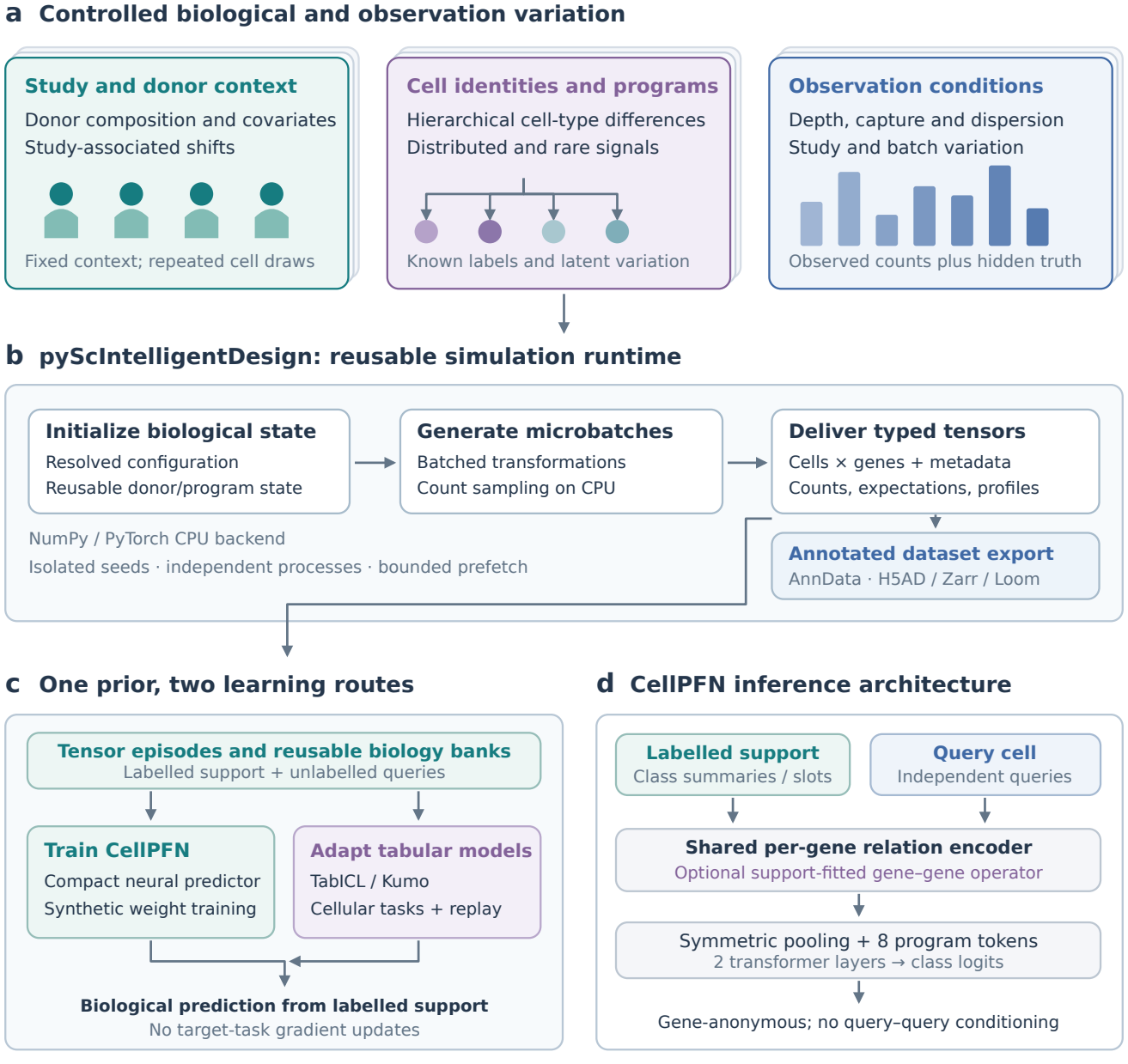
a Controlled biological and observation variation
Study and donor context
Donor composition and covariates
Study-associated shifts
Fixed context; repeated cell draws
Cell identities and programs
Hierarchical cell-type differences
Distributed and rare signals
Known labels and latent variation
Observation conditions
Depth, capture and dispersion
Study and batch variation
Observed counts plus hidden truth
b pyScIntelligentDesign: reusable simulation runtime
Initialize biological state
Resolved configuration
Reusable donor/program state
Generate microbatches
Batched transformations
Count sampling on CPU
Deliver typed tensors
Cells × genes + metadata
Counts, expectations, profiles
NumPy / PyTorch CPU backend
Isolated seeds · independent processes · bounded prefetch
Annotated dataset export
AnnData · H5AD / Zarr / Loom
c One prior, two learning routes
Tensor episodes and reusable biology banks
Labelled support + unlabelled queries
Train CellPFN
Compact neural predictor
Synthetic weight training
Adapt tabular models
TabICL / Kumo
Cellular tasks + replay
Biological prediction from labelled support
No target-task gradient updates
d CellPFN inference architecture
Labelled support
Class summaries / slots
Query cell
Independent queries
Shared per-gene relation encoder
Optional support-fitted gene–gene operator
Symmetric pooling + 8 program tokens
2 transformer layers → class logits
Gene-anonymous; no query–query conditioning